SpatialMapper 3D

ENTERPRISE ARCHITECTURE SPECIFICATION & DEPLOYMENT BLUEPRINT

Document Version: 5.0-UNIFIED **Date:** 10 November 2026 **Prepared For:** Project Owner & Development Team **Stack:** HTML5 / WebGL / Three.js / JS ES6

1. Executive Summary & System Purpose

SpatialMapper 3D is a browser-native, hardware-agnostic 3D spatial reconstruction, metric measurement, and surveillance analytics platform. The application operates entirely client-side inside the browser without requiring external backend servers, active LIDAR sensors, or specialized mobile AR toolkits.

> **Core Capability Overview**
> SpatialMapper 3D converts any standard 2D video feed (Live WebRTC Camera, Video File, YouTube Stream, or Online Player) into a real-time spatial analytics workspace capable of estimating ground distance in metres, entity speeds, time to contact, loitering durations, explainable threat risk metrics, and 10,800-vertex 3D point cloud visualizer models.

2. Visual Aesthetics & Soft-Cyberpunk Design Engine

The interface employs a high-impact **Soft Cyberpunk & Cyber-Organic Nature** aesthetic. It balances futuristic glassmorphism with dynamic ambient lighting and stylized colored botanical artwork elements to deliver an intuitive user experience.

UI Element	Color / Visual Spec	Functional Role
Base Background	Deep Obsidian (`#08070F`)	Provides clean contrast for high-density overlays and WebGL canvases.
Glassmorphic Panels	`rgba(22, 18, 38, 0.55)`	Translucent cards with backdrop blur (16px) and subtle violet glowing borders.
Dynamic Ambient Aura	Shift-on-Action Gradient	Background glow transitions dynamically based on the active subsystem in use.
Cyber-Nature Assets	Stylized Neon Foliage SVGs	Abstract, glowing vector botanical accents (no raw stock photographs).

Dynamic Background Shade Shift Modes

- HOME IDLE Deep Indigo & Lavender (`#1E1B4B` to `#3B0764`)
- LIVE VIDEO Cyber Cyan & Electric Turquoise (`#083344` to `#06B6D4`)
- 3D POINT CLOUD Neon Violet & Magenta (`#581C87` to `#D946EF`)
- CALIBRATION Emerald & Electric Mint (`#064E3B` to `#10B981`)
- RISK ALERT Crimson & Amber Pulse (`#450A0A` to `#F59E0B`)

3. Modular Layered Workspace & Navigation

To eliminate visual clutter, the workspace implements a **Canvas-First Layered Architecture**. Tools do not occupy static sidebars by default; instead, they float as stackable, draggable glass cards toggled on-demand via a floating Quick-Dock HUD.

The 4 System View Layers

1. **Layer 1: Home Landing Portal** — High-impact introduction, interactive capability feature cards, and primary workspace launch trigger.
2. **Layer 2: Modular Workspace (Canvas Core)** — Clean 2D/3D main stage viewport with a floating quick-action pill dock (`Sources`, `Calibration`, `Analytics`, `Zones`, `Logs`, `3D Tools`).
3. **Layer 3: System Settings Modal** — Customization settings that do not impact computation (glass opacity, speed units, alert volumes, dynamic background toggles).
4. **Layer 4: Test Lab & Documentation** — YouTube frame capture guidelines, dataset export tools (YOLO/COCO), precision evaluation benchmark records, open-source licenses, and local AI roadmap status.

4. Lazy-Initialization & Event-Driven Core

> **Performance Safeguard Guarantee**
> On initial page load, the system operates in a completely **Dormant State**. No WebGL contexts, canvas animation loops, WebRTC camera streams, or Web Audio contexts consume system memory or CPU cycles until explicitly triggered.

- **WebRTC Stream Initialization:** Commences ONLY when the user selects a video source and clicks `Start Stream`.
- **WebGL Context Allocation:** Compiles ONLY when the `3D Point Cloud` view or tool module is activated.
- **Automatic Resource Teardown:** Switching tabs away from the Workspace immediately freezes active canvas frame buffers and resets background energy shades to Idle.

5. Mathematical, Geometric & Kinematic Core

All metric estimations rely on rigorous mathematical transformations calculated frame-by-frame:

A. 4-Point Planar Floor Homography

Converts 2D pixel coordinates (u, v) into real-world ground floor coordinates (X, Y) in metres using a 3x3 homography matrix H built from a 4-point known floor rectangle:

$$[X, Y, 1]^T = H \times [u, v, 1]^T$$

B. Inter-Entity Distance & Spatial Kinematics

Ground Euclidean distance between tracked entities 1 and 2:

$$d_{12} = \sqrt{[(X_2 - X_1)^2 + (Y_2 - Y_1)^2]}$$

Velocity vectors and speed classification bands:

$$v = [\Delta X/\Delta t , \Delta Y/\Delta t] , s = ||v||$$

- **Standing:** < 0.3 m/s | **Walking:** 0.3 – 1.8 m/s | **Jogging:** 1.8 – 3.5 m/s | **Running:** > 3.5 m/s (Capped at 12 m/s).

C. Closing Speed & Time to Contact (TTC)

For converging velocity vectors between two entities:

```
TTC = ||r₁₂|| / ||v_rel|| (Triggered when TTC < 2.0 seconds)
```

D. 3D Raycasting Vector Ruler

Measures true 3D Euclidean spatial distance between user-clicked vertices in the point cloud:

```
d₃D = √[ (x₂ - x₁)² + (y₂ - y₁)² + (z₂ - z₁)² ]
```

E. Cryptographic Event Hash Chaining

Guarantees data integrity for export records using sequential SHA-256 hash chaining:

```
H_n = SHA256( H_n-1 + Timestamp + EventData )
```

6. Complete Unified Master Prompt (For AI Agents)

Below is the complete multi-phase master prompt containing all project requirements, mathematical specifications, visual aesthetics, and architectural rules. Copy and paste this directly into any AI coding environment to build or update the project:

```
# SPATIALMAPPER 3D: MASTER UNIFIED MULTI-PHASE SPECIFICATION Version: 5.0-ENTERPRISE-UNIFIED
Role: Principal Computer Vision Architect & Soft-Cyberpunk UI Designer 1. DESIGN ENGINE:
Soft Cyberpunk aesthetic. Deep Obsidian background (`#08070F`), glassmorphic panels
(`rgba(22, 18, 38, 0.55)`), glowing borders (`rgba(168, 85, 247, 0.25)`). Dynamic background
shade transitions (Indigo, Cyber Cyan, Neon Violet, Emerald, Crimson). Abstract glowing
digital foliage vector accents. 2. WORKSPACE LAYOUT: Fixed Top Bar (Home, Workspace,
Settings, Test Lab). Canvas-First Workspace with floating Quick-Dock HUD (`Sources`,
`Calibration`, `Analytics`, `Zones`, `Logs`, `3D Tools`). Tools pop open as draggable,
stackable floating glass cards. 3. LAZY ENGINE: Zero computation on page load. WebRTC,
WebGL, and differencing loops initialize only on explicit user click. Switching tabs freezes
processing. 4. KINEMATICS & MATH: - Homography: [X,Y,1]ᵀ = H × [u,v,1]ᵀ - Distance: d = √[
(X₂-X₁)² + (Y₂-Y₁)² ] - Speed Classes: Standing (<0.3m/s), Walking (0.3-1.8m/s), Jogging
(1.8-3.5m/s), Running (>3.5m/s). - Time to Contact: TTC = ||r₁₂|| / ||v_rel|| - 3D
Raycasting Ruler: d₃D = √[ (x₂-x₁)² + (y₂-y₁)² + (z₂-z₁)² ] - SHA-256 Hash Chain: H_n =
SHA256( H_n-1 + Timestamp + EventData ) 5. EXPORTS & SAFETY: Export 3D clouds as
PLY/OBJ/JSON. Display legal notices: "Automated spatial estimates. Requires human
verification."
```

7. GitHub Repository Structure & AI Studio Migration

To convert your project into a modular repository for zero-cost static deployment via **GitHub Pages**, use the prompt below in **Google AI Studio (Gemini 1.5 Pro)**:

Target Repository File Tree

```
spatialmapper-3d/ ├── index.html # Core HTML5 entry point & navigation structure ├── README.md # Complete repository documentation & deployment guide ├── LICENSE # MIT License ├── css/ | ├── main.css # Soft Cyberpunk glassmorphic styling & CSS variables | └── dynamic-themes.css # Dynamic background shade-shift rules ├── js/ | ├── main.js # App initialization & lazy-loading entry | ├── nav-manager.js # 4-Layer view switcher (Home, Workspace, Settings, About) | ├── theme-engine.js # Dynamic background shade switcher & nature renderer | ├── camera-stream.js # WebRTC camera, YouTube capture & video player logic | ├── homography-math.js # 4-Point floor calibration matrix solver | ├── kinematics.js # Speed vectors, distance, TTC & loitering evaluation | ├── risk-engine.js # Explainable risk meter & alert triggers | ├── hash-logger.js # SHA-256 cryptographic event hash chain manager | └── 3d-cloud.js # Three.js 10,800-vertex point cloud & raycasting ruler └── assets/svg/cyber-foliage.svg
```

Google AI Studio Prompt (Copy-Paste)

```
# GOOGLE AI STUDIO PROMPT: CONVERT SPATIALMAPPER 3D TO GITHUB REPOSITORY Refactor the single-file SpatialMapper 3D application into a clean, modular repository ready for GitHub Pages hosting. Requirements: 1. Target Stack: Standalone HTML5, CSS3, ES6 JS Modules, Three.js (CDN), Canvas API. 2. Direct Deployment: Zero build step required; ready for static GitHub Pages hosting. 3. Modular Structure: Separate code cleanly into `index.html`, `css/main.css`, `css/dynamic-themes.css`, and modular JS files in `js/` (`main.js`, `nav-manager.js`, `theme-engine.js`, `camera-stream.js`, `homography-math.js`, `kinematics.js`, `risk-engine.js`, `hash-logger.js`, `3d-cloud.js`). 4. Aesthetic Preserved: Full Soft-Cyberpunk glassmorphic design, dynamic background color shifts, and floating workspace quick-dock. 5. Lazy Initialization: All camera streams, canvas loops, and WebGL contexts must remain strictly uninitialized until triggered by user interaction. Output the complete, un-truncated code for every file in the directory structure.
```

8. Legal Compliance & Developer Checklist

> **Legal & Privacy Safeguards**
> SpatialMapper 3D operates with privacy by design. All video processing and coordinate math occur locally in client memory. No face recognition is performed, and no video data is transmitted to external servers. Ensure CCTV warning signage is posted in compliance with local rules (GDPR / India DPDP Act 2023).

Recommended Step-by-Step Execution Plan

1. **Run AI Studio Migration Prompt:** Generate the modular repository files using the prompt in Section 7.
2. **Initialize GitHub Repository:** Create a new repository named `spatialmapper-3d` on GitHub and push the generated files.
3. **Enable GitHub Pages:** Go to Repository Settings → Pages → Select `main` branch → Click Save. Your website will be live instantly at `https://<username>.github.io/spatialmapper-3d/`.
4. **Local AI Expansion:** Integrate local Python/ONNX models (YOLO object detection, weapon flags) when running in a desktop/server setup.